1. Which of the following is not a postulate of Dalton's atomic theory ?

(A) All the atoms of a given element are identical in all respect i.e. mass, shape, size etc.

(B) Matter is made up of very small indivisible particles called atom.

(C) Atoms of different element can have same atomic mass.

(D) Atoms cannot be created or destroyed by any chemical process or physical process.

2. 1 amu is equal to :

(A) 1.66×10^{-24} kg
(B) $\frac{1}{8}$ times the mass of one ${}^{16}_{8}O$ -atom

(C) $\frac{1}{2}$ times the mass of one ${}^{1}_{1}H$-atom
(D) $\frac{1}{12}$ times the mass of one ${}^{12}_{6}C$ -atom

3. Which of the following parameter/s is/are unitless ?

(A) RAM (B) GAM (C) Molar mass (D) GMM

4. Mass of 3 atoms of element X is 51 u. Its GAM is **:**

(A) 51 u (B) 17 g (C) 17 u (D) 51 g

5. How many nucleons are present in 2 atoms of an element which has atomic mass 2u ?

(A) 2 (B) 1 (C) 4 (D) 3

6. The ratio of mass of a molecule of X_2 to that of silver atom is 8 : 27. If the molar mass of Ag is 108 g, identify the element X.

(A) ^{14}C (B) ^{15}N (C) ^{16}O (D) ^{14}N

7. The mass of 10 molecules of H_2SO_4 is :

(A) 980 amu
(B) 980 g

(C) $980 \times 1.66 \times 10^{-27}$ kg
(D) Both (A) and (C)

8. Which of the following is correct

(A) GMM = molecular mass in gram
(B) 1 mole = N_A molecules = 6.023×10^{23} molecules

(C) GAM = mass of N_A atoms
(D) all of these

9. What is mass (approx) in gram of an atom with atomic mass 200 amu.

(A) 200 g (B) 3.32×10^{-22} g (C) 3.32×10^{-25} g (D) 0.005 g

1. Total magnitude of charge of 1 mole electron will be approximately :

(A) 19300 C (B) 18500 C (C) 96500 C (D) 9000 C

2. How many moles of atoms are there in one atom

(A) 1 (B) 6.023×10^{23} (C) 1.66×10^{-24} (D) 22.4

3. Number of sulphur atoms present in 0.1 mole S_8 :

(A) 6.023×10^{23} (B) 6.023×10^{24} (C) 48.18×10^{23} (D) 48.18×10^{22}

4. The total number of protons in 63 mg HNO_3 is :

(A) 6.02×10^{20} (B) 1.92×10^{22} (C) 10^{-3} (D) 1

5. Select incorrect matching

(A) $3N_A$ atoms of Ne = 60 g (B) 4 g-atoms of S = 128 g

(C) $6N_A$ molecules of O_3 = 288 g (D) 2 mole H_2 molecules = 2 g

6. Which sample contains the largest number of atoms

(A) 1 mg CH_4 (B) 1 mg N_2 (C) 1 mg Na (D) 1 mg water

7. The least number of molecules are contained in

(A) 2 g Hydrogen (B) 8 g oxygen (C) 4 g Nitrogen (D) 16 g CO_2

8. 1 mole of C_2H_6 contains

(A) N_A molecular of C_2H_6 (B) 24 g of carbon

(C) 3.6×10^{24} H-atoms (D) 18 protons

1. Total number of electrons in 6.2 mg NO_3^- will be :

(A) 16×10^{-4} N_A (B) 32×10^{-4} N_A (C) 48×10^{-4} N_A (D) 80×10^{-4} N_A

2. A sample of D_2 has same mass as that of 10^6 molecules of H_2. The number of molecules of D_2 in the sample is:

(A) 10^6 (B) 5×10^6 (C) 5×10^5 (D) 10^{-6}

3. Find the average molar mass of a mixture of gases containing 48 g O_2 and 0.5 mole H_2.

(A) 49 g (B) 24.5 g (C) 22.5 g (D) 45 g

4. A sample contains 20% by mole of ^{17}O and remaining ^{16}O. The average atomic mass of this sample is :

(A) 16.2 (B) 16.8 (C) 16.4 (D) none of these

5. The largest numbers of molecules are in

(A) 54 g N_2O_5 (B) 28 g CO (C) 36 g H_2O (D) 46 g C_2H_5OH

6. The number of grams of H_2SO_4 in 0.25 mole of it, will be

(A) 0.245 (B) 2.45 (C) 24.5 (D) 49.0

7. The number of atoms in 22 g of carbon dioxide is :

(A) $3 \times 6.02 \times 10^{23}$ (B) $4.5 \times 6.02 \times 10^{23}$

(C) $1.5 \times 6.02 \times 10^{23}$ (D) $2.5 \times 6.02 \times 10^{23}$

1. The temperature at which the volume of ideal gas is zero, is

(A) 0 °C (B) 0 K (C) 0 °F (D) All of these

2. For an ideal gas, the temperature of one mole per litre is given by

(A) 273 K (B) 273°C (C) 298K (D) none of these

3. The volume occupied by 8.8 g CO_2 at 31 °C and 1 atm pressure is

(A) 5 ml (B) 0.2 L (C) 5 L (D) 2 L

4. The temperature of 4.0 moles of an ideal gas occupying 5 L at 3.32 bar will be

(A) 50 K (B) 0.50 K (C) 5×10^2 K (D) 25 K

5. The pressure of a mixture 4 g of O_2 and 2 g H_2 confined in a bulb of 1 litre at 0°C is

(A) 25.215 atm (B) 31.205 atm (C) 45.215 atm (D) 15.210 atm

6. Fixed amount of an ideal gas will have maximun volume when

(A) P = 0.5 atm, T = 600 K (B) P = 2 atm, T = 150 K

(C) P = 1 atm, T = 300 K (D) P = 1 atm, T = 500 K

7. 112 cm^3 of hydrogen gas at STP contains

(A) 0.01 mole (B) 0.5 mole (C) 0.005 mole (D) 0.02 mole

8. 2.8 g of a gas at STP occupies a volume of 2.24 L, the gas can be

(A) O_2 (B) CO (C) NO_2 (D) CO_2

ANSWER KEY

DDT – 01

1.	(C)	**2.**	(D)	**3.**	(A)	**4.**	(B)	**5.**	(C)
6.	(C)	**7.**	(D)	**8.**	(D)	**9.**	(B)		

DDT – 02

1.	(C)	**2.**	(C)	**3.**	(D)	**4.**	(B)	**5.**	(D)
6.	(A)	**7.**	(C)	**8.**	(ABC)				

DDT – 03

1.	(B)	**2.**	(C)	**3.**	(B)	**4.**	(A)	**5.**	(C)
6.	(C)	**7.**	(C)						

DDT – 04

1.	(B)	**2.**	(D)	**3.**	(C)	**4.**	(A)	**5.**	(A)
6.	(A)	**7.**	(C)	**8.**	(B)				